PARHAM KHODADI
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EDUCATION

San Diego State University (SDSU) **B.S. Aerospace Engineering (Minor in Physics)** **08/2023 – 05/2026**
Santa Monica College (SMC) **A.A. in General Sciences** **06/2020 – 07/2023**

EXPERIENCE

Instructional Student Assistant | SDSU College of Engineering **01/2025 – Present**

- Provide academic support for AE320 (Astrodynamics), on topics like two-body orbital mechanics on Keplerian orbits and orbital transfers.
- Provide academic support for AE280 (Differential Equations), reinforcing topics such as Laplace transforms, Fourier series, and linear algebra in engineering contexts through tutoring, preparation, and coordination with faculty.

Guidance Navigation & Control Researcher | SDSU SPACE Lab **11/2023 – 08/2024**

- Programmed DJI Tello EDU drones using djitellopy (Python) for autonomous QR code navigation and string-based command execution.
- Applied OpenCV to extract and decode QR codes, then estimated drone-to-code distance with a pixel-scaling method (±0.5 in accuracy).
- Designed QR-coded indoor navigation paths; drone detected, approached, and followed sequences of embedded movement commands.
- Integrated machine learning-based landing pad detection and tested basic object avoidance using monocular SLAM (ORB-SLAM3).
- Adapted lab environment by applying textured surfaces to improve LiDAR-based altitude control in low-feature terrain.

Avionics and Propulsion Engineer | SDSU Rocket Project **08/2023 – 07/2024**

- Designed and 3D printed custom mounts using SOLIDWORKS for Pressure Transducers, Thermocouples, and Solenoids interfacing with National Instruments DAQ for static fire test data acquisition.
- Assembled Avionics Ground System by soldering and wiring sensors to NI DAQ and developing bulkhead panels for sensor and power interfaces.
- Developed thermal-resistant hardware enclosures for electronics and interface modules using ASA/PLA filaments on Bambu Lab 3D printers.

Researched optimal inlet-to-throat ratios for ethanol-fed Venturi flow meters in Electric Turbopump testing; drafted Arduino code to measure pressure differentials with T200 series sensors before project shift.

Recreation Aide | City of Beverly Hills **04/2019 – 07/2023**

- Led structured activities for children ages 4–14 in camp programs: sports, educational games, arts and crafts, and academic support.
- Adapted during COVID19 to develop remote programming and assisted with recreation operations, including tennis court management.

PROGRAMS

Mission Concept Academy (NASA L'SPACE) **05/2022 – 09/2022**

- Calculated net heat transfer across rover systems and designed thermally optimized housing for Martian rock samples and electronics.
- Selected materials for passive thermal control and developed a dust-proof thermal dissipation panel based on mission constraints.
- Done modules in Siemens NX, Risk Management, Project Management, Systems Engineering, and Heat Transfer.
- Collaborated with a 12-member interdisciplinary team to design a Mars rover mission targeting lava tube exploration.

NASA Proposal Writing & Evaluation Experience (L'SPACE) **01/2022 – 05/2022**

- Developed a manufacturing and prototyping workflow for 2DPA-1 polymer (MIT-discovered material) in space applications.
- Built a year-long Gantt-style timeline to synchronize engineering, research, and testing phases into a cohesive project schedule.
- Participated in NASA-style proposal evaluation, applying official scoring criteria to competing project submissions.
- Supported proposal design using 2DPA-1 in space applications, submitting to NASA for funding consideration.

NASA AFRC Engineering Design Challenge (NCAS Mission 3) **07/2022**

- Led the Engineering sub-team during a NASA Armstrong center program developing wheelchair-accessible seating for eVTOL systems.
- Designed a fold-flat seating mechanism to enable autonomous docking for wheelchairs, integrating mechanical and automation features.
- Shadowed NASA professionals—including a principal investigator and a Pathways intern—and toured advanced eVTOL research facilities.

COURSEWORK

A E 460A + 460B – Spacecraft Design | SDSU

- Developing the conceptual and preliminary design of a Solar Gravitational Lens spacecraft mission: covering mission definition, payload integration, systems architecture, and community-impact considerations.
- Role: Flight Dynamics Systems Engineer – Planning the mission architecture and flight trajectory of the Spacecraft.

A E 303 – Experimental Aerodynamics | SDSU

- Conducted subsonic and supersonic wind tunnel experiments, including airfoil testing, full aircraft model measurements, turbulence sphere studies, and Schlieren/PIV flow visualization.
- Applied uncertainty analysis, statistical methods, and MATLAB to process aerodynamic force, moment, and pressure data.
- Wrote technical reports presenting experimental methods, results, and comparisons to theory and computational tools such as XFOIL.

PUBLICATIONS

P. Khodadi, A. Cook, Y. Tobita, and Dr. X. Liu, *AIAA SDSU Student Branch History*, San Diego State University, 2025. (in-progress, will present at SciTech 2026)

PERSONAL PROJECTS

Atmospheric Magnetoplasmadynamic (MPD) Ion Thruster built using a 1s Lipo battery, DC transformer, nickel strips, and an empty soda can.

YouTube channel "SciPK" where I explain science and engineering topics through video essays.

EXTRACURRICULAR ACTIVITIES

- As **Outreach Officer** of **American Institute of Aeronautics and Astronautics SDSU branch** I organize and lead company tours, speaker panels, and museum events to connect aerospace Engineering students with industry professionals. I also wrote the history paper found in the publications section.
- **Membership Officer** of **Persian Student Association** at SDSU.

SKILLS

Software

- C/C++
  - Unreal Engine
  - Arduino
  - STM32 HAL
- Python
  - OpenCV
  - Numpy
  - Matplotlib
  - Threading
- CAD (SOLIDWORKS and NX)
- MATLAB
- AGI Systems Tool Kit (STK)
- JMARS
- Adobe Premiere Pro
- Adobe Photoshop

Hardware

- Data Acquisition (DAQ)
- Soldering
- Wiring
- 3D Printing

Other Technical or Soft-Skills

- Public Speaking
- LaTeX (via Overleaf)
- Proposal Evaluation
- Mission Concept Design
- Control System Design
- Orbital mechanics
- Research - General
- Teamwork
- Bilingual: Persian, English